HK IMO Training 2023–2024
Problem Set 20
For 9 March 2024

Problem 77. Let S be a set of positive integers with at least two elements. Let $P(x)$ be a polynomial with integer coefficients. Suppose that for any $a, b \in S$, we have $a + b \in S$ and $\gcd(P(a), P(b)) = 1$. Prove that $P(x)$ is a constant polynomial.

Problem 78. Let $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n, c_1, c_2, \ldots, c_n$ be real numbers. Prove that

$$\sum_{\text{cyc}} \sqrt{\sum_{i=1}^{n} (3a_i - b_i - c_i)^2} \geqslant \sum_{\text{cyc}} \sqrt{\sum_{i=1}^{n} a_i^2}.$$

Problem 79. Let $n \geqslant 3$. We colour all edges of a complete graph K_n using n colours such that all colours are used and the edges of the same colour form a single odd cycle. Prove that there exists an odd cycle such that all edges in this cycle have different colours.

Problem 80. Let P be a point inside $\triangle ABC$. Let O_1, O_2, O_3 be the circumcentres of $\triangle APB$, $\triangle BPC$, $\triangle CPA$ respectively. The circumcircles of $\triangle ABC$ and $\triangle O_1O_2O_3$ intersect at two points X and Y. Let Q be the reflection of P in the line XY. Prove that $\angle BAP = \angle CAQ$.